



Original research

Support moment distribution during the back squat at different depths and loads in recreationally trained females

William Goodman^{a,*}, Victoria Flores^c, Joshua A. Cotter^b, David Graham^d, James Becker^a^a Department of Food Systems, Nutrition, & Kinesiology, Montana State University, USA^b Department of Kinesiology, California State University, Long Beach, USA^c Department of Kinesiology and Nutrition, University of Illinois Chicago, USA^d School of Health Sciences and Social Work, Griffith University Gold Coast, AU

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ABSTRACT

Objectives: The squat is used in athletic and clinical settings. However, the coordination of the lower extremity during the lift is not well understood. The purpose was to compare the peak moments of the lower extremity joints at three squat depths (above parallel, parallel and full) and three squat loads (unloaded, 50 % 1 repetition maximum, and 85 % of depth specific 1 repetition maximum) and find their contribution to support (M_s).

Design: Nineteen females performed squats in a randomized order.

Methods: Inverse dynamics and Winter's M_s equation were used to calculate peak moments of the hip, knee and ankle and calculate their contribution to M_s ($\alpha < 0.05$).

Results: Peak hip and ankle extensor moments varied with load but not depth and were greatest when using 85 % 1 repetition maximum. Peak knee extensor moments demonstrated a depth by load interaction. Within each depth as load increased so too did peak knee extensor moments and were highest squatting below parallel when loaded. The hip and knee contribution to M_s demonstrated a depth by load interaction while the ankle was only influenced by load. Within each depth as load increased hip contribution increased whereas the knee decreased contribution. When squatting with load the contribution of the hip decreased at below parallel while the knee increased.

Conclusions: To maximize peak hip moments squat with high load and to maximize peak knee moments squat deep with high load; however, depth and load dosages should be taken into consideration based on the status and goals of the individual.

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1. Introduction

The barbell squat is a commonly used strength and conditioning exercise often incorporated into both athletic development and clinical rehabilitation exercise programs.^{1,2} In athletic development settings, the squat is used broadly to develop hypertrophy, power, and strength of the lower body musculature.^{1,2} In clinical programs, the squat is used for rehabilitation to assist in the progression back to full lower body function following injury.³ Importantly, there is no one preferred method by which practitioners perform the squat movement. This has consequences related to user outcomes because modifying squat parameters such as depth and load can influence training effect and lower extremity joint loading.^{1,2}

Previous research has examined how the joints of the lower limb contribute to squat performance and how variation of squat technique

alters joint contributions. This has primarily focused on modification of squat depth and load, and four primary techniques have been used: net joint moment (NJM), relative muscular effort (RME), electromyography (EMG), and support moment distribution (M_s).^{4,5,7} It has been reported that as squat external load increases, net extensor moments, RME, EMG and peak M_s increase^{4,7,12}; however, it is less clear as to the effects of varied depth on lower extremity muscle activity and joint loading.^{10,11,13} While NJM and RME show that the contribution of the knee extensors increases with depth,^{5,6,9} EMG literature shows that knee extensor muscle activity peaks around parallel depth.^{1,2,5} RME findings provide evidence that contributions from the hip extensors increase with increasing depth^{7,8}; however, hip NJM has not been found to increase past parallel depth.⁵ EMG data are mixed as well in that some studies find hip extensor activity, especially the gluteus maximus, to be greatest at deep depths¹⁴ while others have found hip extensor activity to be greatest at shallow depths¹⁵ and still others have found that the hip extensor activity peaks at parallel depths and plateaus thereafter.¹³

* Corresponding author.

E-mail address: wes@likeironstrong.com (W. Goodman).

The different conclusions regarding the interaction of depth and load between different methodologies make it difficult to provide a consensus statement regarding how changing technique, via changes in depth or load, is produced or mediated by the muscles and joints of the lower body. This may be in part because the methods described, although useful, are subject to the limitations inherent within each analytical method. For example, NJM analysis only considers the net of the flexor and extensor moments and resolves the problem to a single moment expressed at each joint and does not consider the coordination of the lower extremity muscles throughout the movement. RME is a relative contribution of the torque produced around a joint during a task compared to the torque produced in a maximum voluntary contraction. This does not consider the contribution of each joint to the entire lift, but rather, each joint to its own maximum force producing capabilities. Lastly, EMG measures if a muscle is excited and how much it is excited, but it does not measure how much force is produced.

A methodological approach that overcomes many of the aforementioned limitations is the support moment (M_s) which is the sum of the extensor moment of the knee (M_k), extensor moment of the ankle (M_a), and extensor moment of the hip (M_h).¹⁶ M_s is useful because it provides insight into lower extremity coordination by measuring the relative contribution of each joint to the M_s and how that relative contribution changes across activities. Previous research investigating the contribution to peak M_s with increasing loads at a fixed depth of 45 % leg length reported that the ankle contribution increased with loading to 75 % of 3 repetition maximum (RM), the knee contribution decreased when increasing load, and the hip contribution increased with loading to 75 % of 3RM.⁴ In all but the lightest conditions, the hip contributed the most, followed by the knee then ankle.^{4,17}

While this provides insight that joint contributions to total lower body support change with load, it is unknown how joint contributions to lower body support change with depth, or with combinations of increased depth and load. Thus, the purpose of this study is to examine how the ankle, knee, and hip joint moment contributions to M_s change while increasing squat loads and depths. We hypothesized that (1) peak M_s , peak M_a , peak M_k , and peak M_h would increase with load but not depth, (2) percent contribution of the ankle would decrease with depth but increase with load, (3) percent contribution of the knee would increase with depth but not load, and (4) percent contribution of the hip would increase with both depth and load.

2. Methods

A within-subjects experimental design was used to investigate the effects of squatting to different depths with different percentages of 1RM. Participants were evaluated on two separate occasions. During the first session, participants filled out an informed consent and the Physical Activity Readiness Questionnaire (PAR-Q). Depth-specific 1RMs were then obtained for the above parallel (AP), parallel (P) and full (F) depth squats. Depths were defined as follows: AP was measured with a goniometer at 90° of knee flexion, P was reached when the top of the thighs was parallel to the floor, and F was reached when the hamstrings came into complete contact with the back of the calves. A field laser and piezo light switch was used to help establish depth and cue the ascent of the squat (Banner Engineering Corp., Minneapolis, MN).⁹ A brief pause was implemented before the squat ascent. An individualized 10-minute warmup, typical of each participant's usual routine, was implemented prior to testing 1RM for each depth. National Strength and Conditioning Association procedures for 1RM testing were used until the participant could not ascend from the bottom of the squat. During the 1RM testing, loud verbal encouragement and at least 1–2 min of rest were allowed between squat attempts. The stance width and foot position of each participant were recorded with tape and used for later trials.

Within 24 h to 1 week later, participants returned to the laboratory for the second session. During the second session, motion capture data

was collected while two squat repetitions were performed at each of the three different depths with the three different load combinations. The load conditions included unloaded, 50 % 1RM, and 85 % 1RM, and were specific to each depth. A total of nine trials were performed with the order of trials randomized across participants. The specific squat protocol including the 1RM attempt has been elaborated in detail in previous literature.⁹ Whole body kinematics were recorded using a 12-camera motion capture system (Qualisys Inc., Gothenburg, Sweden) sampling at 250 Hz while ground reaction forces were recorded from two force plates (Model 600900-10, Bertec Corporation, Columbus, OH) sampling at 1000 Hz. A 64-channel analog-to-digital board was used to synchronize force and kinematic data.

Retroreflective markers were placed bilaterally on the following bony landmarks: anterior superior iliac spine, posterior superior iliac spine, medial and lateral femoral epicondyles, medial and lateral malleoli, posterior calcaneus, and head of the 2nd metatarsal. Additional tracking markers were placed on the iliac crests, with clusters of four markers on a plastic shell on the lateral thigh and shank, and over the base of the 5th metatarsal.

Nineteen female participants (age: 25.05 ± 5.76 years; height: 1.58 ± 0.08 m; body mass: 62.51 ± 10.17 kg) with an average of 3.84 ± 2.59 years of squatting experience participated in this study. Participants were excluded from participation if they responded “yes” to any questions on the PAR-Q. Competitive, elite lifters and individuals with a history of surgery or lower limb ligamentous injury were also excluded from the study. Inclusion criteria for the female participants included an age range of 18 to 60 years old with a minimum of one year of squat training experience. Visual assessment of the squat was also used to verify that participants could squat to full depth with no pain while keeping heels on the ground and a neutral lumbar spine. The protocol of this study was carried out in accordance with the Declaration of Helsinki, approved by the University Institutional Review Board, and all participants provided consent prior to the study.

Raw kinematic and kinetic data were exported to Visual 3D (C-Motion, Inc., Rockville, MD) where they were filtered using 4th order zero-lag low pass Butterworth filters with cutoff frequencies of 8 Hz and 20 Hz, respectively. Joint angles were calculated using an XYZ Cardan rotation sequence corresponding to flexion/extension, ab/adduction, and axial rotation, respectively. Peak knee joint angles on each trial were then calculated. Net joint moments at the ankle, knee, and hip were calculated using standard inverse dynamics techniques, and expressed as internal joint moments. Total support moment was calculated as described by Winter¹⁶:

$$M_s = M_k + M_a + M_h \quad (1)$$

where M_s is the support moment, M_k is the knee extensor moment, M_a is the ankle plantar flexor moment, and M_h is the hip extensor moment. For each trial the peak joint moments, peak M_s , and the percentage contribution from each joint to peak M_s were calculated.

Participant 1RMs at each depth were analyzed using a one-way repeated measures analysis of variance (ANOVA). For all other dependent variables, values from the two squat repetitions in each condition were averaged and differences between conditions were evaluated using 3×3 (depth $\times$ load) repeated measures ANOVAs. An omnibus alpha level of <0.05 was used to indicate statistical significance, and if appropriate, pairwise comparisons were conducted using a Bonferroni correction. Effect sizes (d , difference in means divided by average standard deviation and η^2 sum of squares (SS) associated with each condition divided by the sum of SS of each condition and its SS error variance) were calculated to aid in interpretation of results, with a d of 0.2, 0.5, and 0.8 and a η^2 of 0.01, 0.06, and 0.14 indicating small, moderate, and large effects, respectively.¹⁸ All statistical analyses were performed using Statistical Package for the Social Sciences v. 26 (IBM SPSS Statistics Inc., Chicago, IL, USA).

3. Results

One repetition maximum was significantly different between depths ($F_{1,14, 20.47} = 10.26$, $p = 0.003$). Values were greatest at AP (72.37 ± 19.33 kg) compared to P (63.16 ± 17.42 kg, $p = 0.017$, $d = 0.50$) or F (58.85 ± 18.52 kg, $p = 0.011$, $d = 0.71$) depths. Values were also significantly greater at P than F depths ($p = 0.048$, $d = 0.24$).

Both depth ($F_{1,48, 26.957} = 225.003$, $p < 0.01$, $\eta^2 = 0.926$) and load ($F_{2, 36} = 5.522$, $p < 0.01$, $\eta^2 = 0.235$) had significant main effects on peak knee flexion angle. Participants squatted deeper at the F ($127.41 \pm 10.26^\circ$) depth than in either P ($103.53 \pm 6.58^\circ$, $p < 0.001$, $d = 2.83$) or AP ($87.48 \pm 7.41^\circ$, $p < 0.001$, $d = 4.81$) depths. Squats at P depth were also deeper than AP depths ($p < 0.001$, $d = 2.29$). The effect of load tended toward participants squatting deeper at P when unloaded ($107.47 \pm 8.38^\circ$, $p = 0.055$, $d = 0.39$) and when using 50 % 1RM ($106.76 \pm 7.45^\circ$, $p = 0.038$, $d = 0.32$) compared to 85 % 1RM ($104.19 \pm 8.44^\circ$), however, this was not statistically significant, and neither was the effect of load on depth at AP or F.

Peak M_s differed with load ($F_{1,113, 20.031} = 248.212$, $p < 0.001$, $\eta^2 = 0.932$) but not with squat depth ($F_{2, 36} = 2.492$, $p = 0.0907$, $\eta^2 = 0.122$). Peak M_s was higher when using 85 % 1RM loads than either 50 % 1RM ($p < 0.001$, $d = 1.08$) or unloaded ($p < 0.001$, $d = 5.97$, Table 1). Peak M_s was also higher with 50 % 1RM than when unloaded ($p < 0.001$, $d = 2.43$, Table 1). Peak hip extensor moments also varied with load ($F_{1,154, 20.764} = 167.038$, $p < 0.001$, $\eta^2 = 0.903$) but not with depth ($F_{2, 36} = 2.162$, $p = 0.130$, $\eta^2 = 0.107$). Peak hip extensor moments were higher when using 85 % 1RM loads than when using either 50 % 1RM ($p < 0.001$, $d = 1.08$) or when unloaded ($p < 0.001$, $d = 3.29$). Peak hip extensor moments were also higher when using 50 % 1RM than when unloaded ($p < 0.001$, $d = 2.35$, Table 1).

Peak knee extensor moments demonstrated a significant depth by load interaction ($F_{2,04, 36.427} = 5.687$, $p = 0.007$, $\eta^2 = 0.240$). Within each depth, as load increased, so too did peak knee extensor moments (Table 1). Within the unloaded condition, there were no differences in peak knee extensor moments between the AP and P depths ($p = 1.000$, $d = 0.04$), the P and F depths ($p = 0.162$, $d = 0.17$), or the AP and F depths ($p = 0.138$, $d = 0.21$). Within the 50 % 1RM condition, peak knee extensor moments were higher in the F depth than in either the P ($p = 0.007$, $d = 0.47$) or AP depths ($p = 0.031$, $d = 0.44$). Finally, in the 85 % 1RM condition, peak knee extensor moments were higher in the F depth than the P depth ($p = 0.009$, $d = 0.73$), but not different between AP and F depths ($p = 0.122$, $d = 0.43$, Table 1).

At the ankle, peak plantar flexor moments varied with load ($F_{1,203, 21.661} = 197.819$, $p < 0.001$, $\eta^2 = 0.917$) but not depth ($F_{2, 36} = 1.138$, $p = 0.332$, $\eta^2 = 0.059$). Peak ankle plantar flexor moments were higher when using 85 % 1RM than when using either 50 % 1RM ($p < 0.001$, $d = 1.28$) or when unloaded ($p < 0.001$,

$d = 3.29$). Peak plantar flexor moments were also higher when using 50 % 1RM than when unloaded ($p < 0.001$, $d = 2.49$, Table 1).

The percentage each joint contributed to the M_s during the squat is shown in Figs. 1 and 2. At peak M_s , the percent contribution from the hip joint demonstrated a depth by load interaction ($F_{2,418, 43.529} = 3.024$, $p = 0.05$, $\eta^2 = 0.144$, Fig. 1). At AP depth, the hip contributed less to peak M_s when squatting unloaded than when using 50 % 1RM ($p < 0.001$, $d = 1.15$) or 85 % 1RM ($p < 0.001$, $d = 1.47$, Fig. 1). When squatting to P depths, the contribution of the hip to peak M_s was lower when unloaded than when using 50 % 1RM ($p < 0.001$, $d = 1.09$) or 85 % 1RM ($p < 0.001$, $d = 1.71$, Fig. 1). Hip contributions to peak M_s at P depth were also lower when using 50 % 1RM than when using 85 % 1RM ($p = 0.003$, $d = 0.59$, Fig. 1). At F depths, results were similar to AP depths, with hip contributions to peak M_s being lower when unloaded than when using 50 % 1RM ($p < 0.001$, $d = 1.03$) or 85 % 1RM ($p < 0.001$, $d = 1.21$), but not different between 50 % 1RM and 85 % 1RM ($p = 0.098$, $d = 0.31$, Fig. 1).

In the unloaded conditions, there were no differences in hip contributions to peak M_s when squatting to different depths (Fig. 2). When using 50 % 1RM, hip contribution to peak M_s was not different between AP and P ($p = 1.00$, $d = 0.06$) or AP and F ($p = 0.229$, $d = 0.35$); however, hip contribution was less at F compared to P ($p = 0.004$, $d = 0.43$, Fig. 2). Similarly, when using 85 % 1RM, hip contribution to peak M_s was not different between AP and P ($p = 0.625$, $d = 0.16$) or AP and F ($p = 0.294$, $d = 0.40$); however, hip contribution was less at F compared to P ($p = 0.029$, $d = 0.61$, Fig. 2).

Percent knee contribution to M_s had a significant depth by load interaction ($F_{2,551, 45.913} = 3.682$, $p < 0.024$, $\eta^2 = 0.170$). At each increasing depth and with each increasing load, knee contribution decreased ($p < 0.001$); however, there was no difference in knee contribution when squatting to F depth when increasing load from 50 % to 85 % 1RM ($p = 0.079$, $d = 0.45$, Fig. 1). When squatting without load, knee contribution was greater at AP compared to P ($p = 0.017$, $d = 0.52$, Fig. 2). When squatting with 85 % 1RM, knee contribution was greater at F than P depths ($p = 0.033$, $d = 0.76$, Fig. 2).

Percent ankle contribution to M_s showed a main effect for load ($F_{1,435, 25.836} = 26.143$, $p < 0.01$, $\eta^2 = 0.592$). Percent ankle contribution increased from unloaded to 50 % 1RM ($p < 0.001$, $d = 0.90$) and unloaded to 85 % 1RM ($p < 0.001$, $d = 1.31$), but did not differ between 50 % and 85 % 1RM ($p = 0.066$, $d = 0.38$, Figs. 1 and 2, respectively).

4. Discussion

This study investigated the effects of load and depth on peak lower extremity moments and contribution of the ankle, knee, and hip to total M_s . In partial support of our hypothesis, the lower extremity

Table 1

Mean and standard deviations for peak total support, hip, knee, and ankle moments at the different depth and load combinations.

		Above parallel	Parallel	Below parallel	Load mean
Peak total support moment (N m/kg)	Unloaded	113.56 $\pm$ 24.74	125.08 $\pm$ 23.12	124.71 $\pm$ 23.49	121.11 $\pm$ 6.54*
	50 % 1RM	190.07 $\pm$ 32.07	192.95 $\pm$ 38.01	196.08 $\pm$ 35.76	193.04 $\pm$ 3.01*
	85 % 1RM	235.33 $\pm$ 45.51	236.25 $\pm$ 46.84	242.81 $\pm$ 50.64	238.13 $\pm$ 4.08*
	Depth mean	179.65 $\pm$ 61.55	184.76 $\pm$ 56.04	187.87 $\pm$ 59.48	
Peak hip moment (N m/kg)	Unloaded	41.75 $\pm$ 14.08	49.18 $\pm$ 11.55	48.95 $\pm$ 10.26	46.63 $\pm$ 4.22*
	50 % 1RM	87.07 $\pm$ 23.06	89.96 $\pm$ 25.05	88.50 $\pm$ 22.84	88.51 $\pm$ 1.44*
	85 % 1RM	116.24 $\pm$ 31.09	119.79 $\pm$ 31.10	117.45 $\pm$ 31.70	117.83 $\pm$ 1.80*
	Depth mean	81.69 $\pm$ 37.54	86.31 $\pm$ 35.45	84.96 $\pm$ 34.39	
Peak knee moment (N m/kg)	Unloaded	59.42 $\pm$ 14.32	59.99 $\pm$ 13.71	62.38 $\pm$ 14.38	60.60 $\pm$ 1.57
	50 % 1RM	71.67 $\pm$ 15.41	71.03 $\pm$ 15.67	79.06 $\pm$ 18.21 [#]	73.92 $\pm$ 4.46
	85 % 1RM	80.93 $\pm$ 18.37	75.80 $\pm$ 16.51	89.33 $\pm$ 20.78 [#]	82.02 $\pm$ 6.83
	Depth mean	70.67 $\pm$ 10.78	68.93 $\pm$ 8.11	76.83 $\pm$ 13.59	
Peak ankle moment (N m/kg)	Unloaded	18.88 $\pm$ 5.54	22.34 $\pm$ 6.65	21.77 $\pm$ 6.81	21.00 $\pm$ 1.86*
	50 % 1RM	37.25 $\pm$ 7.63	39.61 $\pm$ 9.14	41.23 $\pm$ 8.47	39.36 $\pm$ 2.00*
	85 % 1RM	53.92 $\pm$ 13.35	52.87 $\pm$ 13.69	52.80 $\pm$ 12.71	53.20 $\pm$ 0.63*
	Depth mean	36.68 $\pm$ 17.53	38.28 $\pm$ 15.31	38.60 $\pm$ 15.68	

Note: * indicates main effect of load while [#] indicated depth by load interaction, at the $p < 0.05$ level. See Results text for descriptions of main effects and interactions.

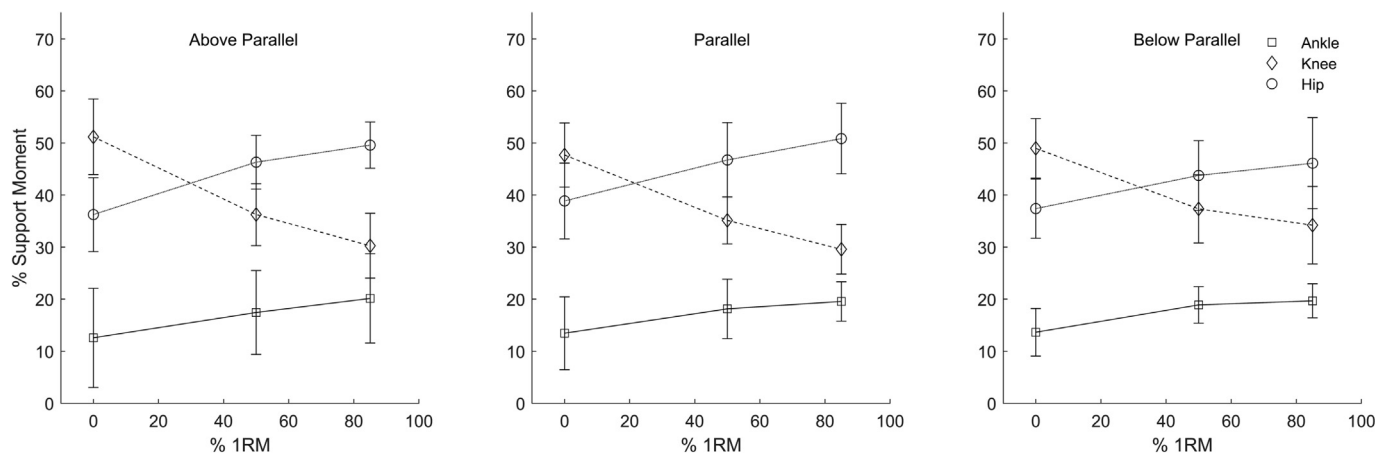


Fig. 1. Joint contributions (percent peak support moment) for each of the three squat depths when squatting with different percentages 1RM (mean $\pm$ SD).

peak moments are more affected by increasing external loads than by increasing squat depth. In opposition to our hypothesis, increasing depth did not significantly change the contribution of the ankle. In partial support of our hypothesis, the hip was influenced by depth and load; however, in opposition to our hypothesis, the F depth squat resulted in the lowest hip contributions and squatting with greater than 50% 1RM tended not to further increase hip contributions. In opposition to our hypothesis, the contribution of the knee decreased from AP to P, yet, in support of our hypothesis, increased from P to F depths. Since this occurred in tandem with a decrease in hip contributions from P to F depths, at deep depths when using heavy loads, the knee contributes more to the squat while the hip contributes less. Furthermore, contribution of the ankle and the hip increases with loading while the knee decreases contribution, indicating that a peak joint moment can increase in absolute terms while its contribution to support can decrease.

The results of the current study align with previous research indicating that increasing load increases peak lower extremity moments during the barbell back squat^{3,4,9} and that peak knee extensor moments are greatest at full depths.^{5,9,10} Our findings also align with previous studies evaluating contribution to M_s with increased load as, similar to Flanagan et al. (2008), we observed that the contribution of the knee decreased with increasing load and the contribution of the hip and ankle increased with increasing load.⁴ However, our results differ from previous literature in that we did not find peak hip extensor moments to be influenced by depth.^{5,7} Our results may differ from previous literature because we used depth-specific 1RMs while these studies used the 1RM from the deepest depth to establish test loads. Maximum squat strength is joint angle specific.^{9,10} If full depth squat 1RMs are used to dictate shallower depth loads, it is unlikely to represent the appropriate test load. Therefore, it is possible that previous literature

found hip moments to increase with depth because the load prescribed was based on the full squat 1RM and was not sufficient to challenge the hip musculature at shallower depths.

Direct comparison to studies which have used RME to assess how muscles contribute to squat performance is difficult, as the different approaches lead to different conclusions. Byranton et al. (2012) reported that hip contributions to the squat increased with both squat depth and load while knee contributions were more affected by depth than load.⁷ They concluded that heavy, deep squats should be performed to train the hip extensors while light, deep squats should be performed to train the knee extensors.⁷ In contrast, the current study observed that hip contributions to squat performance were more affected by load than depth, suggesting that hip extensors can be trained with heavy loads at relatively shallow depths. However, the knee contribution to squat performance decreased with higher loads but increased with depth.

Several studies using EMG amplitude to assess muscle function during squats have reported that EMG amplitude increases with increasing squat load.^{19–21} However, there is disagreement in the literature regarding how squat depth influences muscle activity. Contreras et al. reported no differences in hip muscle activity between parallel and deep squats.¹³ In contrast, da Silva et al.¹⁵ observed that above parallel squats resulted in greater hip muscle activity than deep squats and Caterisano et al.¹⁴ observed lower hip muscle activity during partial and parallel squats compared to deep squats.¹³ There is less discrepancy at the knee as Contreras et al.,¹³ da Silva et al.,¹⁵ and Caterisano et al.¹⁴ all reported no difference in quadriceps muscle activity between partial, parallel, and full depth squats.

The current study conceptually supports some of the previous EMG findings as we observed that peak M_s and peak hip, knee and ankle extensor moments all increased with load and that peak hip extensor

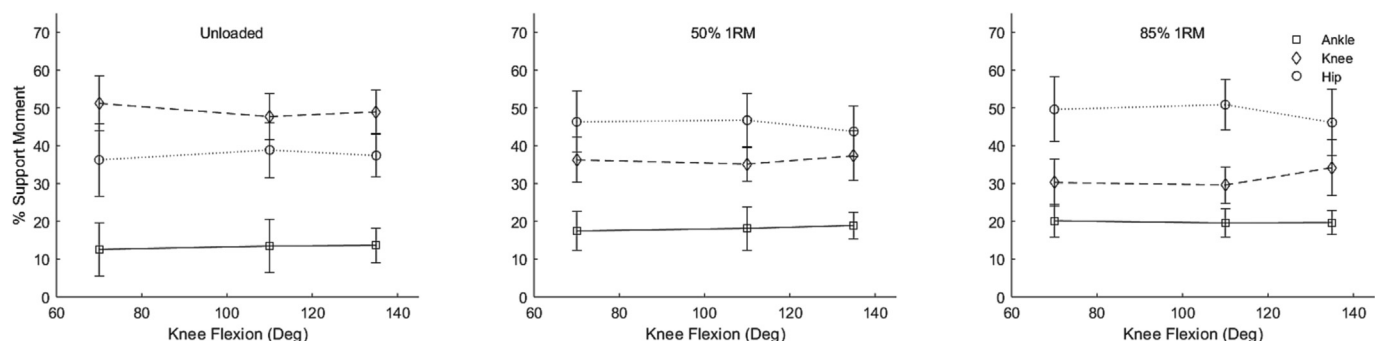


Fig. 2. Joint contributions (percent peak support moment) for each of the three squat loads when squatting to different depths (mean $\pm$ SD).

moments were not different with increasing squat depth. However, direct comparison between results from peak moments and EMG activity is not possible as the peak moments represent the peak net moment about the joint and could be influenced by co-contraction. No changes in quadriceps muscle activity with increasing squat depth, if occurring in combination with decreased hamstring muscle activity, could result in higher net knee extensor moments. This might explain the increased peak knee extensor moments with deep, heavy squats, despite no differences in EMG. Furthermore, previous studies are not consistent with the methodology used to established squat loads and the level of training experience among participants. Previous studies have used a 1RM from the deepest depth to prescribe squat loads across all depths,^{5,7} while others have used depth-specific 1RMs to prescribe squat loads.^{9,10} These differences in methodologies are likely to yield different results because squat strength is depth specific.¹⁰ The level of squat experience is also likely to influence the results. Previous literature indicates that co-activation of antagonist muscles is higher in novice lifters than those that are more experienced.²² This could influence the joint moments and EMG findings between studies.

There are several limitations to our study which need to be considered when interpreting the results. First, bar position was not controlled. Previous research has suggested that there are kinematic differences between the high bar and low bar back squat techniques.^{23,24} While most participants used a high bar position (16 high bar and 3 low bar), we cannot rule out that differences in bar position influenced the results. However, if we required everyone to use one bar position compared to the other one this may have also influenced the results due to the inexperience with the untrained position. Similarly, while each participant's stance width was the same for both testing days, it was not controlled across participants. Gluteus maximus activity has been shown to increase with increasing stance width,^{20,21} which could influence the hip moment values. However, these two studies used extreme stance widths not representative of normal squatting technique and that were not used in the current study. Our study investigated recreationally trained females. These finding may change based on the sex and training status of the participants and should therefore be replicated with stronger, more experienced females and males. Finally, shoe type was not controlled. While many of the participants wore tennis sneakers, one did wear squatting shoes. These shoes reduce forward trunk lean and increase plantar flexion compared to wearing running shoes, and potentially contribute to greater knee extensor activity.²⁵ Squat shoes have also been suggested to reduce ankle dorsiflexion, increase knee flexion, decrease trunk lean and increase knee moments in an unloaded condition compared to an athletic shoe.²⁶

In conclusion, this study investigated the effects of load and squat depth on peak moments of the lower extremity joints and their contribution to M_s during the squat. Lower extremity joint moments and contributions to the M_s were more affected by load than depth. Peak moments at the hip, knee, and ankle increase with load, but not necessarily with depth. However, the contribution to M_s of the knee and hip changes with depth and load while the ankle is only influenced by load. The findings from this study indicate that although the peak extensor moment of a joint may increase, the contribution to overall squat performance as determined by the M_s can increase, decrease, or stay the same.

5. Conclusion

- If the goal is to train the hip extensors, we recommend focusing on increased loads.
- If the goal is to train the knee extensors, we recommend training to deep depths with as large a load as is reasonable for the participant.
- Peak moments increase with increasing loads and knee moments are greatest when squatting to deep depths. When rehabilitating, appropriate load and depth dosages should be taken into consideration based on the status of the individual. For example, if there is a hip

issue, consider using light loads and if there is a knee issue consider using light loads to shallow depths for a period of time.

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Confirmation of ethical compliance

This research was approved by the Institutional Review Board of California State University Long Beach and in accordance with the Declaration of Helsinki.

CRediT authorship contribution statement

William Goodman: Conceptualization, Methodology, Software, Formal analysis, Writing. **Victoria Flores:** Investigation, Resources, Writing – Review and editing. **Joshua Cotter:** Investigation, Resources, Writing – Review and editing. **David Graham:** Supervision, Writing. **James Becker:** Supervision, Writing, Formal analysis, Software, Conceptualization, Methodology.

Declaration of interest statement

There are no financial or personal relationships with other people or organizations that could inappropriately influence our work.

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